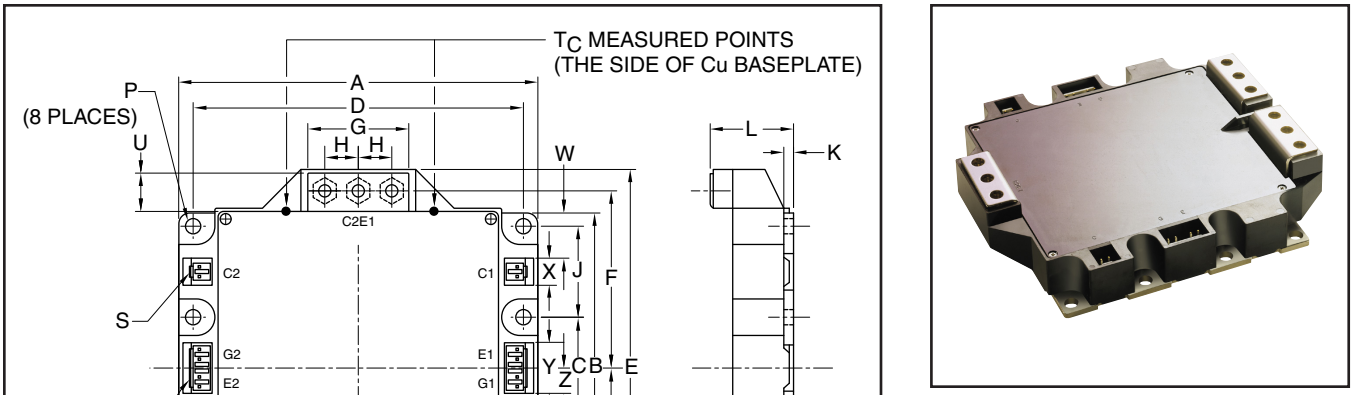
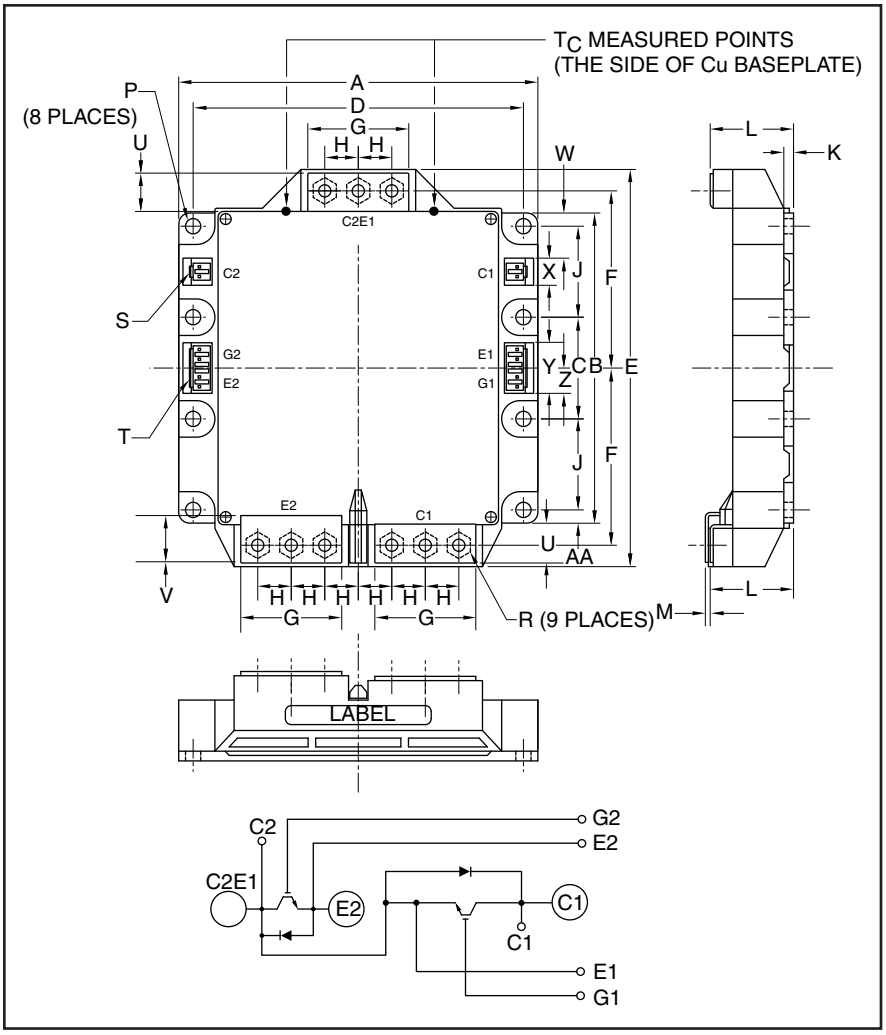
CM1400DU-24NF

Powerex, Inc., 173 Pavilion Lane, Youngwood, Pennsylvania 15697 (724) 925-7272
www.pwrx.com

Mega Power Dual IGBTMOD™
1400 Amperes/1200 Volts





Outline Drawing and Circuit Diagram

Dimensions	Inches	Millimeters
A	5.91	150.0
B	5.10	129.5
C	1.67±0.01	42.5±0.25
D	5.41±0.01	137.5±0.25
E	6.54	166.0
F	2.91±0.01	74.0±0.25
G	1.65	42.0
H	0.55	14.0
J	1.50±0.01	38.0±0.25
K	0.16	4.0

Housing Type (J.S.T. MFG. CO. LTD)
S = VHR-2N
T = VHR-5N

Dimensions	Inches	Millimeters
L	1.36 +0.04/-0.02	34.6 +1.0/-0.5
M	0.075±0.08	1.9±0.2
P	0.26	6.5
R	M6 Metric	M6
U	0.62	15.7
V	0.71	18.0
W	0.75	19.0
X	0.43	11.0
Y	0.83	21.0
Z	0.41	10.5
AA	0.22	5.5

Description:
Powerex IGBTMOD™ Modules are designed for use in switching two IGBT applications. Each module consists of a half-bridge configuration, with each transistor having a reverse-connected super-fast recovery free-wheel diode. All components and interconnects are isolated from the heat sinking baseplate, offering simplified system assembly and thermal management.

Features:
- ☐ Low Drive Power
- ☐ Low $V_{CE(sat)}$
- ☐ Discrete Super-Fast Recovery Free-Wheel Diode
- ☐ Isolated Baseplate for Easy Heat Sinking

Applications:
- ☐ High Power UPS
- ☐ Large Motor Drives
- ☐ Utility Interface Inverters

Ordering Information:
Example: Select the complete module number you desire from the table - i.e. CM1400DU-24NF is a 1200V (V_{CES}), 1400 Ampere Dual IGBTMOD Power Module.

Type	Current Rating Amperes	V_{CES} Volts (x 50)
CM	1400	24

CM1400DU-24NF
Mega Power Dual IGBTMOD™
1400 Amperes/1200 Volts

Maximum Ratings, T_j = 25°C unless otherwise specified

Ratings	Symbol	Ratings	Units
Collector-Emitter Voltage (G-E SHORT)	V_{CES}	1200	Volts
Gate-Emitter Voltage (C-E SHORT)	V_{GES}	±20	Volts
Collector Current DC ($T_{C'}$ = 94°C)*5	I_C	1400	Amperes
Peak Collector Current (Pulse)*2	I_{CM}	2800	Amperes
Emitter Current (T_C = 25°C)	I_E*1	1400	Amperes
Peak Emitter Current (Pulse)*2	I_{EM}*1	2800	Amperes
Maximum Collector Dissipation (T_C = 25°C)	P_C*3	3900	Watts
Junction Temperature	T_j	-40 to 150	°C
Storage Temperature*4	T_{stg}	-40 to 125	°C
Isolation Voltage (Terminals to Baseplate, f = 60Hz, AC 1 min.)	V_{iso}	2500	Volts
Mounting Torque, M6 Mounting Screws	–	40	in-lb
Mounting Torque, M6 Main Terminal Screw	–	40	in-lb
Weight (Typical)	–	1400	Grams

Electrical Characteristics, T_j = 25°C unless otherwise specified

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
Collector-Cutoff Current	I_{CES}	V_{CE} = V_{CES}, V_{GE} = 0V	–	–	1	mA
Gate-Emitter Threshold Voltage	$V_{GE(th)}$	I_C = 140mA, V_{CE} = 10V	6	7	8	Volts
Gate Leakage Current	I_{GES}	±V_{GE} = V_{GES}, V_{CE} = 0V	–	–	1.5	µA
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$	I_C = 1400A, V_{GE} = 15V, T_j = 25°C*4	–	1.8	2.5	Volts
(Without Lead Resistance)	(Chip)	I_C = 1400A, V_{GE} = 15V, T_j = 125°C*4	–	2.0	–	Volts
Module Lead Resistance	$R_{(lead)}$	I_C = 1400A, Terminal-Chip	–	0.286	–	mΩ
Input Capacitance	C_{ies}	V_{CE} = 10V, V_{GE} = 0V	–	–	220	nF
Output Capacitance	C_{oes}		–	–	25	nF
Reverse Transfer Capacitance	C_{res}		–	–	4.7	nF
Total Gate Charge	Q_G	V_{CC} = 600V, I_C = 1400A, V_{GE} = 15V	–	7200	–	nC
Turn-on Delay Time	$t_{d(on)}$	V_{CC} = 600V, I_C = 1400A, V_{GE} = ±15V, R_G = 0.22Ω, Inductive Load, I_E = 1400A	–	–	800	ns
Turn-on Rise Time	t_r		–	–	300	ns
Turn-off Delay Time	$t_{d(off)}$		–	–	1000	ns
Turn-off Fall Time	t_f		–	–	300	ns
Reverse Recovery Time	t_{rr}*1		–	–	700	ns
Reverse Recovery Charge	Q_{rr}*1		–	90	–	µC
Emitter-Collector Voltage	V_{EC}*1	I_E = 1400A, V_{GE} = 0V	–	–	3.2	Volts
(Without Lead Resistance)	(Chip)					
Internal Gate Resistance	r_g	Per Switch	–	0.67	–	Ω

*1 Represent ratings and characteristics of the anti-parallel, emitter-to-collector free wheeling diode (FWDi).
*2 Pulse width and repetition rate should be such that device junction temperature (T_j) does not exceed $T_{j(max)}$ rating.
*3 Junction temperature (T_j) should not increase beyond maximum junction temperature ($T_{j(max)}$) rating.
*4 Pulse width and repetition rate should be such as to cause negligible temperature rise.
*5 Case temperature ($T_{C'}$) measured point is just under the chips. If you use this value, Rth(f-a) should be measured just under the chips.
*8 The operation temperature is restrained by the permission temperature of female connector.

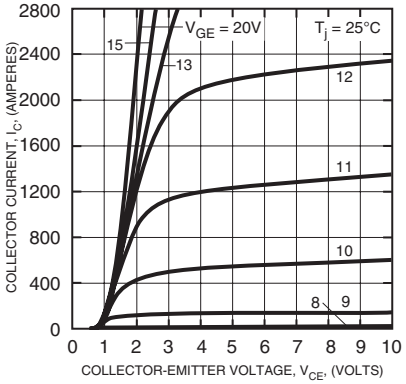
CM1400DU-24NF
Mega Power Dual IGBTMOD™
1400 Amperes/1200 Volts

Thermal and Mechanical Characteristics, T_j = 25°C unless otherwise specified

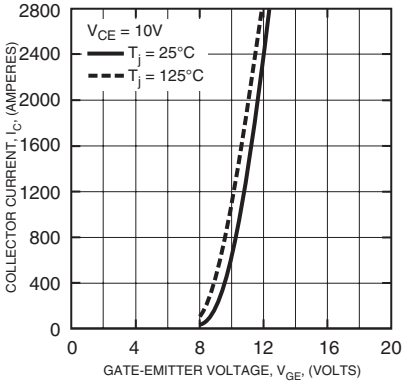
Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
Thermal Resistance, Junction to Case*7	$R_{th(j-c)}Q$	IGBT Part (1/2 Module)	–	–	0.032	K/W
Thermal Resistance, Junction to Case*7	$R_{th(j-c)}D$	FWDi Part (1/2 Module)	–	–	0.053	K/W
Contact Thermal Resistance*6	$R_{th(c-f)}$	Case to Heatsink, Thermal Grease Applied (1/2 Module)	–	0.016	–	K/W
Thermal Resistance, Junction to Case*5	$R_{th(j-c')}Q$	Per IGBT Part, T_C Reference Point Under the Chips	–	–	0.014	K/W
Thermal Resistance, Junction to Case*5	$R_{th(j-c')}D$	Per FWDi Part, T_C Reference Point Under the Chips	–	–	0.023	K/W
External Gate Resistance	R_G		0.22	–	2.2	Ω

*5 Case temperature ($T_{C'}$) measured point is just under the chips. If you use this value, Rth(f-a) should be measured just under the chips.
*6 Typical value is measured by using thermally conductive grease of λ = 0.9 [W/(m • K)].
*7 Case temperature (T_C) measured point is shown in the device dtawing.

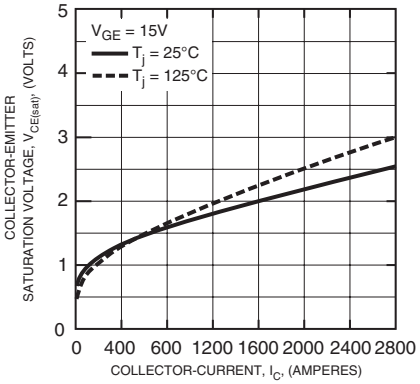
OUTPUT CHARACTERISTICS (TYPICAL)



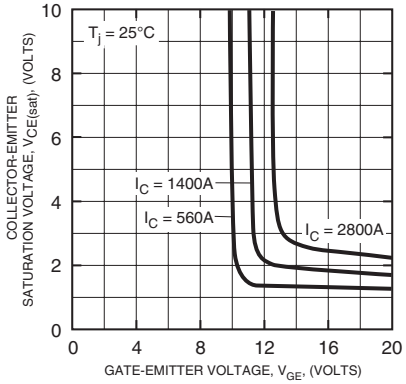
TRANSFER CHARACTERISTICS (TYPICAL)



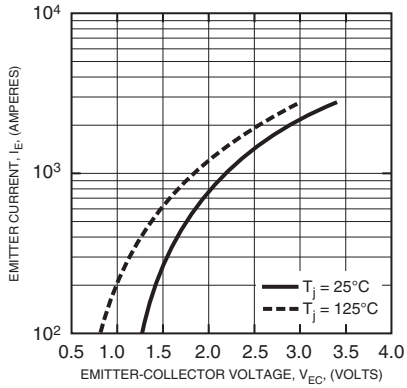
COLLECTOR-EMITTER SATURATION VOLTAGE CHARACTERISTICS (TYPICAL)



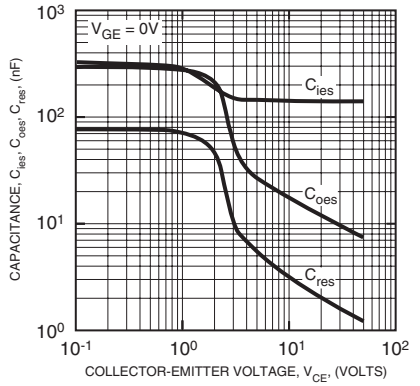
COLLECTOR-EMITTER SATURATION VOLTAGE CHARACTERISTICS (TYPICAL)



FREE-WHEEL DIODE FORWARD CHARACTERISTICS (TYPICAL)

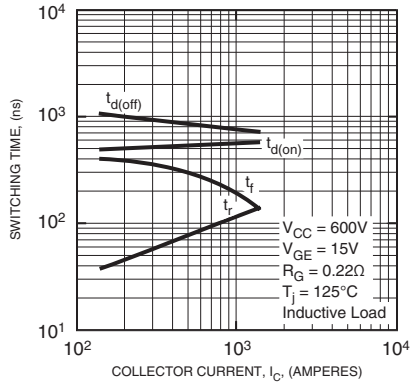


CAPACITANCE VS. V_{CE} (TYPICAL)

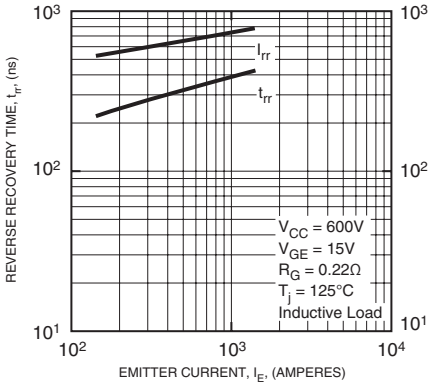


CM1400DU-24NF
Mega Power Dual IGBTMOD™
1400 Amperes/1200 Volts

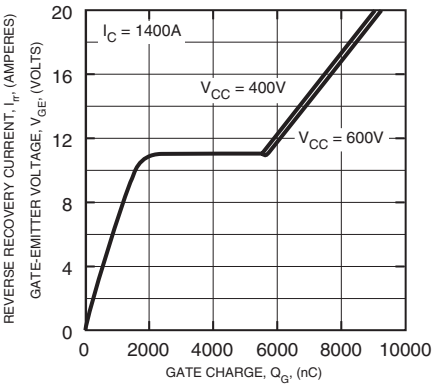
HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)



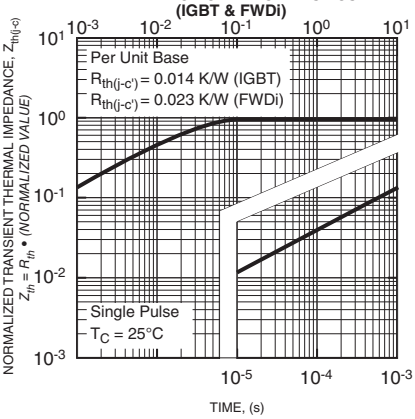
REVERSE RECOVERY CHARACTERISTICS (TYPICAL)



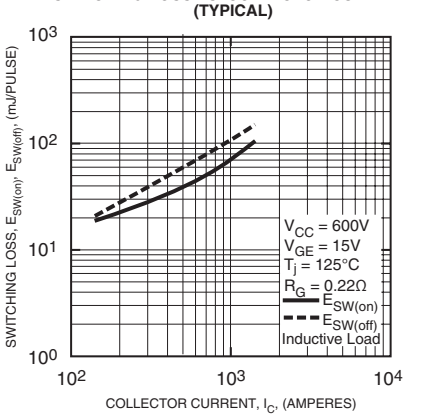
GATE CHARGE, V_{GE}



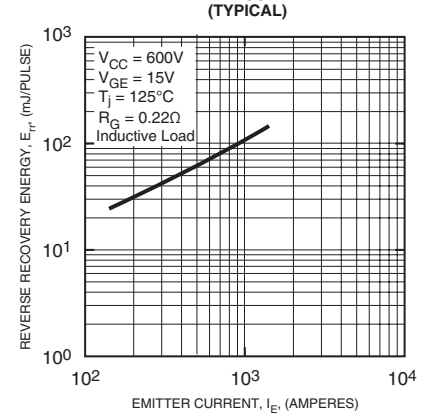
TRANSIENT THERMAL IMPEDANCE CHARACTERISTICS (IGBT & FWDi)



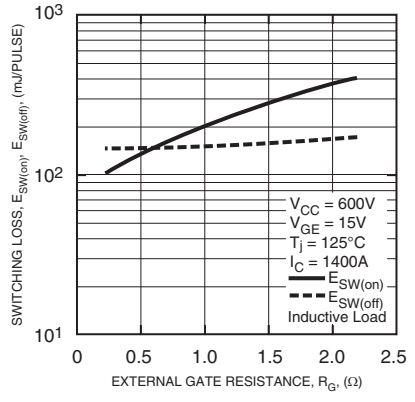
SWITCHING LOSS VS. COLLECTOR CURRENT (TYPICAL)



REVERSE RECOVERY ENERGY VS. EMITTER CURRENT (TYPICAL)



SWITCHING ENERGY VS. EXTERNAL GATE RESISTANCE (TYPICAL)



REVERSE RECOVERY ENERGY VS. EXTERNAL GATE RESISTANCE (TYPICAL)

